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Design and implementation of an AI-based fruit-picking robotic system

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CHAPTER ONE

GENERAL INTRODUCTION

1.1 Project Overview and Background

The integration of artificial intelligence, computer vision, and robotics has created new opportunities for automating agricultural operations. Among these operations, fruit harvesting presents a significant engineering challenge because it requires accurate visual perception, spatial localization, and careful physical interaction with delicate produce. Unlike structured industrial environments, orchards contain irregularly distributed objects, changing illumination, and obstacles that complicate automated harvesting.

This project, entitled "Design and Implementation of an AI-Based Fruit-Picking Robotic System," aims to develop an intelligent robotic system for apple harvesting. The proposed system combines a mobile platform, a robotic arm, a gripper, and a vision system to support fruit detection, localization, and subsequent harvesting operations. The long-term objective is to develop a miniature robotic prototype capable of selectively harvesting ripe apples and transferring them to a collection basket mounted on the robot.

1.2 Problem Statement and Significance

Apple harvesting is particularly challenging because fruits may be partially hidden by leaves and branches, clustered together, or located at different positions within the tree canopy. An automated system must therefore distinguish individual fruits from their surroundings and estimate their spatial positions with sufficient accuracy to support robotic manipulation. In addition, the harvesting mechanism must account for fruit size, weight, and attachment conditions to reduce the risk of damage during picking and collection.

These challenges highlight the need for a robotic system capable of combining reliable fruit perception with accurate spatial localization and controlled manipulation. Establishing these capabilities is important for reducing dependence on manual harvesting and supporting the development of automated agricultural operations.

1.3 Project Objectives

The proposed system aims to support automated apple harvesting through the integration of visual perception, robotic mobility, and fruit manipulation. Its primary functions are to identify apples, determine their spatial positions, and provide the information required for subsequent harvesting operations.

The objectives of the current development phase are:

- Develop a virtual apple orchard and robotic model using ROS 2 and Gazebo.

- Integrate a simulated camera capable of providing color and depth information.
- Detect apples in captured images using a YOLO-based approach.
- Estimate the three-dimensional positions of detected apples using depth information and camera calibration parameters.
- Evaluate detection accuracy, processing speed, and localization performance under different simulated conditions.

The long-term objectives include developing a miniature physical robot capable of identifying ripe apples, approaching and detaching them safely, avoiding obstacles, and transferring harvested fruits into an onboard collection basket.

1.4 Definitions and Key Concepts

The following concepts are fundamental to understanding the proposed system:

Artificial Intelligence (AI): A field concerned with developing computational systems capable of performing tasks associated with intelligent behavior, including learning, recognition, and decision-making.

Computer Vision: A field of computing that develops methods for extracting and interpreting information from digital images and videos.

Robot Operating System 2 (ROS 2): A software framework that provides tools and communication mechanisms for developing and integrating robotic applications.

Gazebo: A robotics simulation environment used to model robots, sensors, and their interactions with virtual environments.

YOLO (You Only Look Once): A family of deep-learning-based object detection models designed to identify and locate objects within images.

RGB-D Camera: A sensing device that provides color images and corresponding depth measurements, supporting the estimation of object distances and spatial positions.

Three-Dimensional Localization: The process of estimating an object's position in three-dimensional space relative to a defined coordinate system.

Robotic Manipulator: A mechanical system consisting of interconnected links and joints designed to position and move an end-effector within a specified workspace.

End-Effector: A device attached to the end of a robotic manipulator to interact with objects, such as a gripper or suction mechanism.

1.5 Project Scope and Technical Requirements

The technical specifications define the essential requirements for developing the proposed intelligent apple-harvesting robotic system. These requirements are derived from the intended harvesting operations, the characteristics of the target fruit, the operating environment, and the findings of the reviewed research.

At this stage, the specifications focus on the functions that each subsystem must perform rather than prescribing specific commercial components or final mechanical dimensions. This approach allows alternative design solutions to be evaluated before the system configuration is finalized.

The requirements are organized into environmental, mechanical, sensing, software, control, and safety considerations. They also distinguish between the simulation-based development planned for the current phase and the physical harvesting operations intended for subsequent implementation.

1.5.1 System Scope and Development Phases

The development process is divided into two phases.

Phase I: Simulation and Perception Development

The first phase is limited to simulation-based development and evaluation of apple detection and three-dimensional localization. The simulated system will include an orchard environment, a robotic model, and a virtual camera capable of providing color and depth information. Physical fruit detachment and collection are outside the scope of this phase.

Phase II: Physical Prototype Development

The second phase involves constructing a miniature robotic prototype capable of interacting with real apples. The intended functions include moving through the operating environment, identifying suitable fruits for harvesting, approaching them safely, detaching them from the tree, and transferring them into a collection basket mounted on the robot.

The physical design and component selection will be determined according to the results of the preliminary studies, the required functions, and the practical constraints of the prototype.

1.5.2 Fruit and Operating Environment Requirements

The proposed system is intended to handle real apples with variations in size, weight, shape, and maturity. These characteristics must be considered when developing the visual perception system and selecting a suitable harvesting mechanism.

The vision system must distinguish apples from surrounding leaves, branches, and other background objects. It should also support the identification of individual fruits when several apples appear close together or partially overlap.

For selective harvesting, the system is intended to distinguish ripe apples from unripe ones. This function is necessary to avoid harvesting fruits that have not reached the desired maturity. The criteria used to determine ripeness will require further investigation and validation.

The operating environment may contain uneven ground, tree trunks, branches, and other obstacles. Consequently, the robotic platform must be capable of maintaining stability during movement, while the robotic arm must operate without causing unnecessary contact with surrounding objects.

Variations in illumination and partial fruit occlusion must also be considered because they can affect the reliability of visual detection and localization.

During the current phase, these environmental conditions will be represented in the simulation to support the evaluation of the perception system.

1.5.3 Mechanical and Functional Requirements

The mechanical structure of the proposed robot consists of a mobile platform, a robotic arm, a fruit-gripping mechanism, and an onboard collection basket. Each component serves a specific function within the harvesting process.

Mobile Platform

The mobile platform provides the structural support required to carry the robotic arm, sensing equipment, control components, and collection basket. It must maintain sufficient stability during movement and harvesting operations.

A suitable locomotion mechanism is required to enable movement through the intended operating environment. Its selection should consider ground conditions, maneuverability, stability, and the additional load resulting from collected fruit.

Robotic Arm

The robotic arm is responsible for positioning the harvesting mechanism near the target apple. It must provide sufficient freedom of movement and reach to access fruits within the intended workspace.

The arm should also support controlled movements to reduce the possibility of collisions with branches, surrounding fruits, or other parts of the robot.

Its configuration will be selected according to the required workspace, expected load, and complexity of the harvesting task.

Fruit-Gripping Mechanism

A suitable end-effector is required to establish controlled contact with the apple and support its detachment from the tree.

The mechanism must accommodate variations in fruit size and shape while limiting excessive pressure that could cause bruising or other damage. The final gripping approach will be selected after evaluating the requirements of safe fruit handling and detachment.

Onboard Collection Basket

A collection basket must be mounted on the robotic platform to receive harvested apples.

The harvesting mechanism must be capable of transferring detached fruits toward the basket and releasing them in a controlled manner. The collection process should minimize the risk of damage caused by impact or uncontrolled dropping.

The basket and its accumulated contents must also be considered when evaluating the stability and load distribution of the mobile platform.

1.5.4 Vision and Sensing Requirements

The vision system is responsible for acquiring the information required to recognize apples and estimate their positions.

A sensing device capable of providing color images is necessary to distinguish fruit from the surrounding environment. Depth information is also required to estimate the distance between the camera and detected apples and to determine their three-dimensional positions.

The perception system must identify individual apples within the camera's field of view and provide their image locations for further processing.

For selective harvesting, an additional image-analysis function will be required to assess fruit maturity according to defined visual characteristics. This function must be evaluated under different illumination and visibility conditions.

The physical prototype will also require information about nearby obstacles to support safe movement and robotic arm operation. The sensing approach should therefore provide sufficient environmental awareness to reduce the risk of collisions.

Sensor selection will depend on the required detection range, field of view, processing capability, and operating conditions.

1.5.5 Software and Control Requirements

The software architecture must support communication and coordination among the perception, localization, simulation, and control components.

During the first phase, ROS 2 will provide the communication framework for exchanging sensor data and processing results. Gazebo will be used to simulate the orchard environment, robotic structure, and sensing operations.

A YOLO-based detection model is planned to identify apples in color images. The detected image locations will be combined with depth information and camera calibration parameters to estimate the three-dimensional coordinates of the target fruits.

Coordinate transformations will be required to express fruit positions relative to the robot's reference frames. Visualization tools will be used to inspect the detection and localization results.

The system should also support the collection of performance measurements, including detection accuracy, processing speed, and localization error.

In the physical implementation phase, the control system will additionally coordinate platform movement, robotic arm positioning, fruit handling, and transfer to the collection basket.

The intended operating modes should allow the system to be monitored and controlled during testing, with appropriate provisions for manual intervention when necessary.

1.5.6 Safety and Reliability Requirements

Safety is an essential consideration because the proposed robot is intended to operate near trees, fruits, and potentially human operators.

The robotic platform and manipulator must be designed to reduce the risk of unintended collisions. Movement should be stopped or adjusted when an obstacle prevents safe operation.

The fruit-handling mechanism must avoid excessive gripping force and uncontrolled release. If a fruit cannot be reached or detached safely, the system should be capable of abandoning the attempt rather than applying potentially damaging actions.

The physical prototype must also include a means of stopping operation in an emergency. Appropriate protective measures will be required to address electrical faults, excessive actuator loads, and loss of communication or control.

During simulation, safety-related conditions can be investigated through collision monitoring, movement constraints, and controlled test scenarios.

1.5.7 Performance Evaluation and Verification

The performance of the proposed system must be evaluated using criteria appropriate to each development phase.

For the simulation phase, the evaluation will focus on the ability to detect apples, estimate their three-dimensional positions, and maintain reliable processing under different simulated conditions.

Detection performance will be assessed using established measures such as precision, recall, and mean average precision. Processing speed will be evaluated in frames per second, while localization accuracy will be determined by comparing estimated fruit positions with reference positions obtained from the simulation environment.

The evaluation should include different fruit arrangements, lighting conditions, and levels of partial occlusion to examine the limitations of the proposed perception approach.

During the physical implementation phase, additional tests will be required to assess movement stability, obstacle avoidance, harvesting success, fruit damage, and the reliability of transferring harvested apples to the onboard basket.

The results of these evaluations will be used to determine whether the system satisfies its intended functional requirements and to identify areas requiring further improvement.

1.6 Project Contribution and Novelty

The proposed project's contribution lies in developing an integrated simulation-based framework for apple detection and three-dimensional localization as an initial step toward robotic harvesting. Rather than introducing a new object detection algorithm, the project aims to combine established technologies within a coordinated robotic development environment.

The main contribution is the planned integration and evaluation of these components in a virtual orchard, providing a basis for investigating perception performance before physical harvesting is introduced. This work is intended to support subsequent development of a miniature robotic prototype rather than claim a new harvesting algorithm or an experimentally validated improvement over existing systems.

1.7 Design Methodology

The project follows a phased development approach. The preliminary work has involved reviewing four relevant research papers and preparing an initial technical specifications document. These activities provide the foundation for identifying the system's functional requirements, examining possible design alternatives, and defining the next implementation steps.

The initial implementation stage will focus on developing a virtual orchard and robotic model using ROS 2 and Gazebo. A simulated RGB-D camera will provide image and depth data for apple detection and spatial localization. A YOLO-based model will be applied to identify apples, and the resulting detections will be combined with depth information to estimate their three-dimensional positions. The system will then be evaluated under different simulated conditions.

The selection of technical solutions will be guided by their suitability for the required functions, compatibility with the simulation environment, and ability to support subsequent physical implementation. Construction of the robotic prototype and testing with real fruit are planned for a later development phase.

1.7.1 Project Development Timeline

The initial stage involved reviewing four scientific papers related to robotic apple harvesting, computer vision, and three-dimensional fruit localization. Based on these studies, a preliminary technical specifications document was prepared to identify the system's functional, mechanical, environmental, and software requirements. This work forms the basis of the first seminar, scheduled for 31 October 2026.

The next stage focuses on developing a simulation environment using ROS 2 and Gazebo. This stage includes creating a virtual apple orchard, modeling the mobile robot and robotic arm, and integrating a simulated camera. A YOLO-based detection model will then be used to identify apples in captured images, while depth information will support the estimation of their three-dimensional positions.

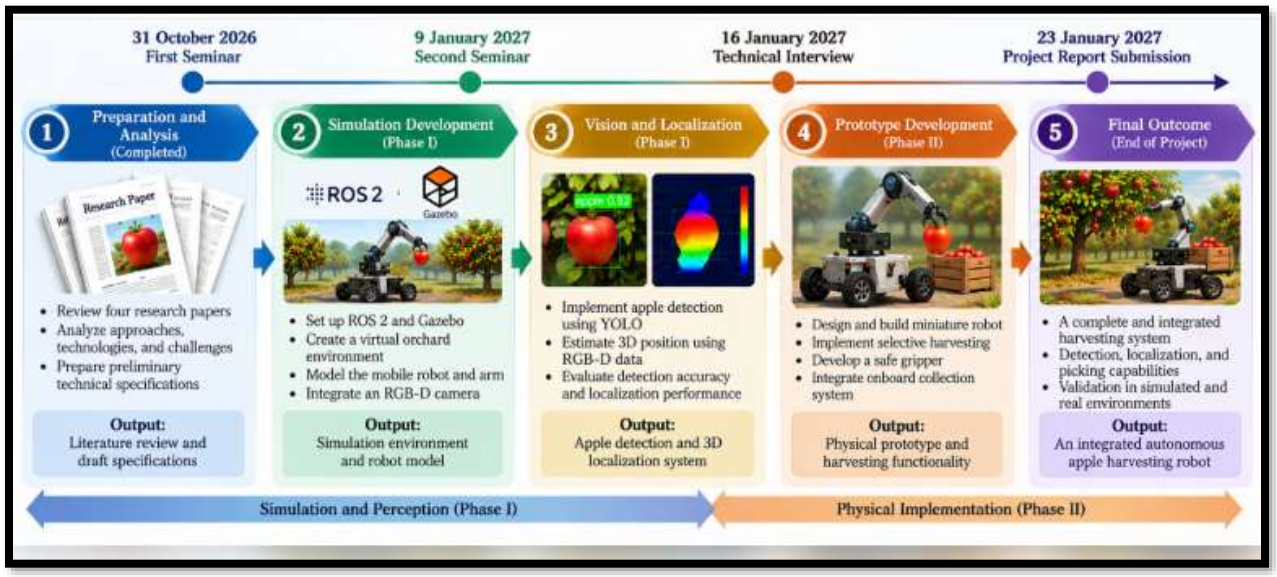
The second seminar, scheduled for 9 January 2027, will present the progress achieved during simulation development and any preliminary experimental results. The technical interview on 16 January 2027 will provide an opportunity to discuss the system architecture, implementation methods, technical challenges, and available performance measurements.

The project report submission is scheduled for 23 January 2027. The report will document the development methodology, completed implementation activities, experimental findings, and identified limitations. The expected outcome of this phase is a simulation-based system capable of detecting apples and estimating their three-dimensional positions, with its performance evaluated under different simulated conditions.

In a subsequent development phase, the project will progress toward constructing a miniature physical robot equipped with a robotic arm, a suitable gripper, and an onboard collection basket. This phase will focus on selective harvesting of ripe apples, safe fruit handling, obstacle avoidance, and testing with real fruit.

The proposed sequence of activities and academic milestones is illustrated in Figure 1.

Figure 1: Project Development Timeline and Scheduled Milestones



1.8 Report Structure

This report is organized into three main chapters. Chapter One presents the project background, problem statement, objectives, technical requirements, and development methodology. Chapter Two reviews relevant research studies, compares existing approaches, identifies the research gap, and discusses the theoretical foundations of the selected technologies. Chapter Three introduces the preliminary system architecture and the main mechanical, sensing, and software subsystems considered for the proposed design.

1.9 Chapter Conclusion

This chapter defined the scope and requirements of the proposed robotic apple-harvesting system and established a phased approach that separates simulation-based perception development from future physical harvesting. The resulting framework provides a basis for examining relevant research and selecting suitable technical solutions, which are addressed in the following chapter.

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Chapter 2

LITERATURE REVIEW AND TECHNICAL SPECIFICATIONS

Robotic fruit harvesting has attracted increasing research interest because of the challenges associated with automating selective picking operations. Unlike conventional industrial

manipulation, harvesting requires robots to identify target fruits within complex natural scenes, estimate their positions, and interact with them while minimizing the risk of damage.

Research in this field has explored different approaches to visual perception, depth estimation, robotic arm design, end-effector mechanisms, and system integration. These approaches vary according to the type of fruit, orchard conditions, sensing technologies, and intended level of automation.

Four research papers were selected for this study to examine complementary aspects of apple-harvesting systems. The first provides a broad review of robotic fruit-picking technologies. The second investigates the design and field evaluation of an automated apple-harvesting robot. The third focuses on apple detection and three-dimensional localization using deep learning and depth information. The fourth examines the integration of robotic harvesting components through ROS 2.

Together, these studies provide a foundation for understanding the technical decisions involved in the proposed system and identifying the challenges that should be considered during its development.

2.1 Automatic Fruit Picking Technology: A Comprehensive Review of Research Advances

Zhang et al. (2024) [1] presented a comprehensive review of technological developments in robotic fruit harvesting. The study examined the main components of automated harvesting systems, including fruit recognition, spatial localization, robotic manipulators, and end-effectors. It also discussed the challenges associated with operating in agricultural environments, particularly variations in illumination, fruit occlusion, irregular fruit distribution, and the risk of mechanical damage during harvesting.

The review highlighted the importance of integrating reliable perception methods with appropriate mechanical designs. It also showed that harvesting performance depends not only on the accuracy of fruit detection but also on the robot's ability to reach and detach fruit safely.

As a review article, the study did not introduce or experimentally validate a new harvesting robot. Nevertheless, it provides a useful foundation for identifying the functional requirements of the proposed system and comparing alternative approaches to robotic arms and gripping mechanisms.

2.2 An Automated Apple Harvesting Robot: From System Design to Field Evaluation

Zhang et al. (2024) [2] developed and evaluated an automated robotic system for apple harvesting under real orchard conditions. The system integrated an apple detection method based on O2RNet, depth sensing, laser-assisted localization, a four-degree-of-freedom robotic arm, and a vacuum-based harvesting mechanism.

The researchers investigated the coordination between visual perception and robotic manipulation to locate and harvest apples. Field experiments demonstrated harvesting success rates of 82.4% in sparse orchard conditions and 65.2% in dense orchard conditions, with an average harvesting time of approximately six seconds per apple.

The difference in performance between orchard conditions illustrates the influence of fruit accessibility and environmental complexity on robotic harvesting. For the proposed project, this study provides practical insights into system integration, fruit localization, and the mechanical requirements of safe harvesting. However, its physical implementation is more advanced than the scope of the current simulation phase.

2.3 Dual-Detector Vision and Depth-Aware Back-Projection for Accurate Apple Detection and 3D Localisation for Robotic Harvesting

Hossain et al. (2026) [3] investigated a vision-based approach for apple detection and three-dimensional localization. The proposed method combined YOLOv8 and Mask R-CNN with RGB-D sensing and depth-aware back-projection to estimate the spatial positions of detected apples.

The study reported a mean average precision of 94.2% for the two-dimensional detection stage, together with a recall of 92.8% and an F1-score of 93.3%. A processing rate of 86.42 frames per second was also reported for the detection stage. These results demonstrate the potential of deep learning and depth information for efficient fruit perception.

The main contribution of this research lies in combining image-based detection with depth-based spatial estimation. However, the study focused on perception rather than the physical execution of harvesting operations.

This research is particularly relevant to the first phase of the proposed project, which aims to detect apples using a YOLO-based model and estimate their three-dimensional positions from simulated RGB-D data.

2.4 Robotic Harvesting of Apples Using ROS2

Ruybalid et al. (2026) [4] presented a robotic apple-harvesting system developed using the Robot Operating System 2 (ROS 2). The system integrated a mobile robotic platform, a robotic manipulator, visual sensing devices, and software components responsible for coordinating perception and harvesting operations.

The perception system incorporated YOLOv5 for apple detection, while stereo and RGB-D cameras supported different stages of fruit localization. ROS 2 provided the communication framework for exchanging information among the system components.

Experimental evaluations reported a harvesting success rate of 100% under laboratory conditions and success rates of 75% and 79% in two field evaluations. These results reflect the additional challenges encountered when transferring robotic harvesting operations from controlled environments to real orchards.

This study demonstrates the importance of software integration and coordinated operation between perception and manipulation components. It is relevant to the proposed system because ROS 2 will be used to connect the simulated camera, detection algorithm, localization module, and robotic

model. Unlike the proposed first phase, however, the study evaluated a physical harvesting robot rather than a Gazebo-based simulation.

2.5 Comparative Analysis of Previous Studies

The reviewed studies address complementary aspects of robotic apple harvesting. Zhang et al. [1] provided a broad understanding of harvesting technologies and their limitations, while Zhang et al. [2] demonstrated the practical integration of perception and manipulation in a physical harvesting system. Hossain et al. [3] concentrated on apple detection and three-dimensional localization, whereas Ruybalid et al. [4] emphasized ROS 2-based integration of robotic harvesting components.

A comparison of these studies indicates that accurate fruit perception is a necessary foundation for automated harvesting, but it does not independently guarantee successful picking. Physical harvesting also depends on fruit accessibility, robotic arm capabilities, gripping mechanisms, and the coordination of system components.

Accordingly, the proposed project adopts a phased approach. The initial phase focuses on developing and evaluating apple detection and three-dimensional localization within a simulated orchard. The subsequent phase will address physical implementation, selective harvesting, obstacle avoidance, and fruit collection.

This approach allows the perception system to be examined before introducing the additional mechanical and operational challenges of a physical prototype.

Chapter 3

Technical Specifications

The technical specifications define the essential requirements for developing the proposed intelligent apple-harvesting robotic system. These requirements are derived from the intended

harvesting operations, the characteristics of the target fruit, the operating environment, and the findings of the reviewed research.

At this stage, the specifications focus on the functions that each subsystem must perform rather than prescribing specific commercial components or final mechanical dimensions. This approach allows alternative design solutions to be evaluated before the system configuration is finalized.

The requirements are organized into environmental, mechanical, sensing, software, control, and safety considerations. They also distinguish between the simulation-based development planned for the current phase and the physical harvesting operations intended for subsequent implementation.

3.1 System Objectives and Scope

The proposed system aims to support automated apple harvesting through the integration of visual perception, robotic mobility, and fruit manipulation. Its primary functions are to identify apples, determine their spatial positions, and provide the information required for subsequent harvesting operations.

The development process is divided into two phases.

Phase I: Simulation and Perception Development

The first phase focuses on establishing a virtual environment in which apple detection and three-dimensional localization can be developed and evaluated. The simulated system will include an orchard environment, a robotic model, and a virtual camera capable of providing color and depth information. The main objective is to investigate the reliability of fruit perception under different environmental conditions.

Phase II: Physical Prototype Development

The second phase involves constructing a miniature robotic prototype capable of interacting with real apples. The intended functions include moving through the operating environment,

identifying suitable fruits for harvesting, approaching them safely, detaching them from the tree, and transferring them into a collection basket mounted on the robot.

The physical design and component selection will be determined according to the results of the preliminary studies, the required functions, and the practical constraints of the prototype.

3.2 Fruit and Operating Environment Requirements

The proposed system is intended to handle real apples with variations in size, weight, shape, and maturity. These characteristics must be considered when developing the visual perception system and selecting a suitable harvesting mechanism.

The vision system must distinguish apples from surrounding leaves, branches, and other background objects. It should also support the identification of individual fruits when several apples appear close together or partially overlap.

For selective harvesting, the system is intended to distinguish ripe apples from unripe ones. This function is necessary to avoid harvesting fruits that have not reached the desired maturity. The criteria used to determine ripeness will require further investigation and validation.

The operating environment may contain uneven ground, tree trunks, branches, and other obstacles. Consequently, the robotic platform must be capable of maintaining stability during movement, while the robotic arm must operate without causing unnecessary contact with surrounding objects.

Variations in illumination and partial fruit occlusion must also be considered because they can affect the reliability of visual detection and localization.

During the current phase, these environmental conditions will be represented in the simulation to support the evaluation of the perception system.

3.3 Mechanical and Functional Requirements

The mechanical structure of the proposed robot consists of a mobile platform, a robotic arm, a fruit-gripping mechanism, and an onboard collection basket. Each component serves a specific function within the harvesting process.

Mobile Platform

The mobile platform provides the structural support required to carry the robotic arm, sensing equipment, control components, and collection basket. It must maintain sufficient stability during movement and harvesting operations.

A suitable locomotion mechanism is required to enable movement through the intended operating environment. Its selection should consider ground conditions, maneuverability, stability, and the additional load resulting from collected fruit.

Robotic Arm

The robotic arm is responsible for positioning the harvesting mechanism near the target apple. It must provide sufficient freedom of movement and reach to access fruits within the intended workspace.

The arm should also support controlled movements to reduce the possibility of collisions with branches, surrounding fruits, or other parts of the robot.

Its configuration will be selected according to the required workspace, expected load, and complexity of the harvesting task.

Fruit-Gripping Mechanism

A suitable end-effector is required to establish controlled contact with the apple and support its detachment from the tree.

The mechanism must accommodate variations in fruit size and shape while limiting excessive pressure that could cause bruising or other damage. The final gripping approach will be selected after evaluating the requirements of safe fruit handling and detachment.

Onboard Collection Basket

A collection basket must be mounted on the robotic platform to receive harvested apples.

The harvesting mechanism must be capable of transferring detached fruits toward the basket and releasing them in a controlled manner. The collection process should minimize the risk of damage caused by impact or uncontrolled dropping.

The basket and its accumulated contents must also be considered when evaluating the stability and load distribution of the mobile platform.

3.4 Vision and Sensing Requirements

The vision system is responsible for acquiring the information required to recognize apples and estimate their positions.

A sensing device capable of providing color images is necessary to distinguish fruit from the surrounding environment. Depth information is also required to estimate the distance between the camera and detected apples and to determine their three-dimensional positions.

The perception system must identify individual apples within the camera's field of view and provide their image locations for further processing.

For selective harvesting, an additional image-analysis function will be required to assess fruit maturity according to defined visual characteristics. This function must be evaluated under different illumination and visibility conditions.

The physical prototype will also require information about nearby obstacles to support safe movement and robotic arm operation. The sensing approach should therefore provide sufficient environmental awareness to reduce the risk of collisions.

Sensor selection will depend on the required detection range, field of view, processing capability, and operating conditions.

3.5 Software and Control Requirements

The software architecture must support communication and coordination among the perception, localization, simulation, and control components.

During the first phase, ROS 2 will provide the communication framework for exchanging sensor data and processing results. Gazebo will be used to simulate the orchard environment, robotic structure, and sensing operations.

A YOLO-based detection model is planned to identify apples in color images. The detected image locations will be combined with depth information and camera calibration parameters to estimate the three-dimensional coordinates of the target fruits.

Coordinate transformations will be required to express fruit positions relative to the robot's reference frames. Visualization tools will be used to inspect the detection and localization results.

The system should also support the collection of performance measurements, including detection accuracy, processing speed, and localization error.

In the physical implementation phase, the control system will additionally coordinate platform movement, robotic arm positioning, fruit handling, and transfer to the collection basket.

The intended operating modes should allow the system to be monitored and controlled during testing, with appropriate provisions for manual intervention when necessary.

3.6 Safety and Reliability Requirements

Safety is an essential consideration because the proposed robot is intended to operate near trees, fruits, and potentially human operators.

The robotic platform and manipulator must be designed to reduce the risk of unintended collisions. Movement should be stopped or adjusted when an obstacle prevents safe operation.

The fruit-handling mechanism must avoid excessive gripping force and uncontrolled release. If a fruit cannot be reached or detached safely, the system should be capable of abandoning the attempt rather than applying potentially damaging actions.

The physical prototype must also include a means of stopping operation in an emergency. Appropriate protective measures will be required to address electrical faults, excessive actuator loads, and loss of communication or control.

During simulation, safety-related conditions can be investigated through collision monitoring, movement constraints, and controlled test scenarios.

3.7 Performance Evaluation and Verification

The performance of the proposed system must be evaluated using criteria appropriate to each development phase.

For the simulation phase, the evaluation will focus on the ability to detect apples, estimate their three-dimensional positions, and maintain reliable processing under different simulated conditions.

Detection performance will be assessed using established measures such as precision, recall, and mean average precision. Processing speed will be evaluated in frames per second, while localization accuracy will be determined by comparing estimated fruit positions with reference positions obtained from the simulation environment.

The evaluation should include different fruit arrangements, lighting conditions, and levels of partial occlusion to examine the limitations of the proposed perception approach.

During the physical implementation phase, additional tests will be required to assess movement stability, obstacle avoidance, harvesting success, fruit damage, and the reliability of transferring harvested apples to the onboard basket.

The results of these evaluations will be used to determine whether the system satisfies its intended functional requirements and to identify areas requiring further improvement.

Table 1Comparison of Selected Research Studies on Robotic Apple Harvesting:

Research Paper	Technical Approach	Main Contribution	Relevance to our Project

Identifying suitable system requirements and design alternatives	Comparison of selective harvesting technologies and their challenges	Review of vision systems, robotic arms, grippers, and harvesting methods	**Automatic Fruit Picking Technology**
Understanding fruit localization, handling, and practical harvesting challenges	Design and field evaluation of an integrated harvesting robot	O2RNet, RGB-D sensing, laser-based localization, 4-DOF arm, suction gripper	**An Automated Apple Harvesting Robot**
Supporting our proposed detection and 3D position estimation pipeline	Apple detection and 3D localization	YOLOv8, Mask R-CNN, RGB-D depth processing	**Dual-Detector Vision and Depth_Aware Back-Prejection**
Guiding software communication and future system integration	Integration of perception and robotic harvesting components	ROS 2, YOLOv5, stereo/RGB-D cameras, robotic arm	**Robotic Harvesting of Apples Using ROS2**

Research Gap and Project Direction

The reviewed studies demonstrate significant progress in robotic apple harvesting, particularly in visual perception, three-dimensional localization, and the integration of robotic components. However, these studies address different aspects of the harvesting process and employ varying levels of system complexity.

For the proposed project, the primary challenge is to establish a reliable perception and localization framework before proceeding to physical harvesting operations. Therefore, the initial development phase will focus on integrating YOLO-based apple detection with RGB-D depth information in a ROS 2 and Gazebo simulation environment.

This approach enables the evaluation of detection and localization performance under controlled conditions while providing a foundation for future integration with robotic manipulation, selective harvesting, and onboard fruit collection.